

Labwork 2: Linear Regression from Scratch

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1 Algorithm Implementation

1.1 Initialization and Objective Function

At first, we initialize random w_0 , w_1 , learning rate r and threshold t :

$$w_0 = 0, w_1 = 1, r = 0,0001, t = 10 \quad (1)$$

To calculate a single loss value, we use the function:

$$f(w_1, w_0, x_i, y_i) = \frac{1}{2}(w_1 x_i + w_0 - y_i)^2 \quad (2)$$

The total loss L for all data points with N samples is:

$$L(w_1, w_0) = \frac{1}{N} \sum_{i=1}^N \frac{1}{2}(w_1 x_i + w_0 - y_i)^2 \quad (3)$$

1.2 Gradient Descent Optimization

The update rules for each iteration are as follows:

$$w_0 := w_0 - r \cdot \frac{\partial L}{\partial w_0} \quad (4)$$

$$w_1 := w_1 - r \cdot \frac{\partial L}{\partial w_1} \quad (5)$$

where r is the learning rate. The partial derivatives (gradients) are calculated as follows:

$$\frac{\partial L}{\partial w_0} = \frac{1}{N} \sum_{i=1}^N (w_1 x_i + w_0 - y_i) \quad (6)$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{N} \sum_{i=1}^N x_i (w_1 x_i + w_0 - y_i) \quad (7)$$

2 Visualization

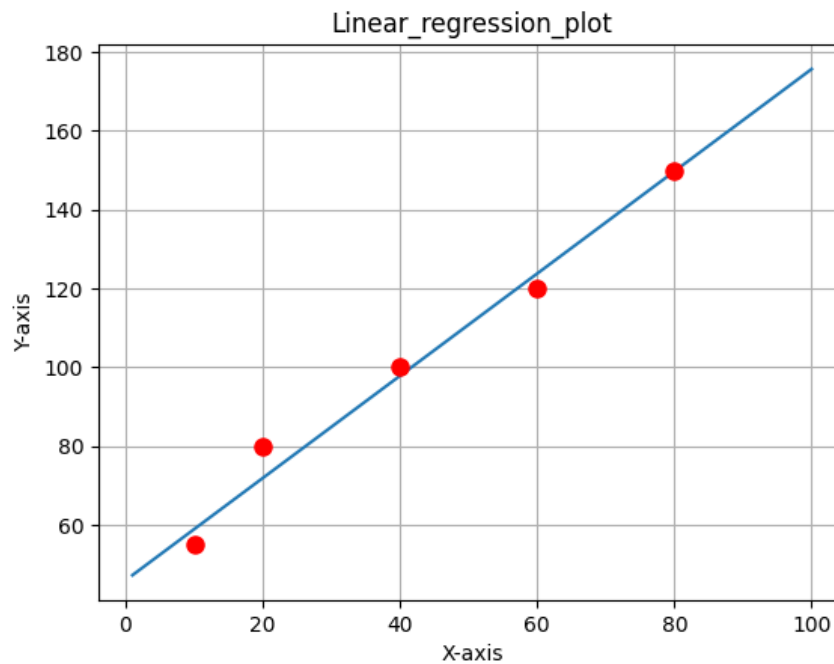


Figure 1: Data and Regression Line.

3 Analysis of Learning Rate (r)

The learning rate is the most critical hyperparameter in this implementation:

- **Small learning rate:** The algorithm takes very small steps. Although stable, it converges very slowly and requires a massive number of iterations to find the minimum value.
- **Optimal learning rate (Good):** The algorithm works efficiently. It moves towards the minimum point steadily and converges to the correct parameters in a reasonable number of iterations.
- **Large learning rate:** The algorithm takes steps that are too large. It may overshoot (jump over) the minimum point, causing the loss value to oscillate rather than strictly decrease.
- **Very large learning rate:** The algorithm completely fails to converge. The loss diverges because the updates move the weights further away from the minimum in every step.